Jordan University of Science and Technology

Chemistry Lab 107

Second semester 2020

Done by:

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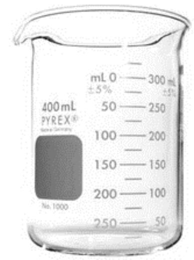

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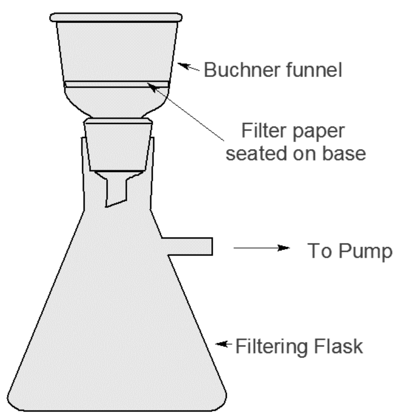
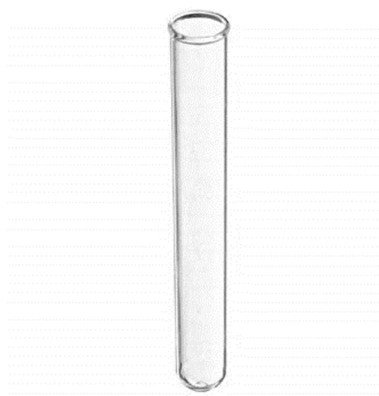
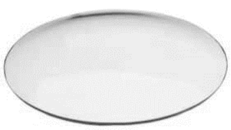
Wire gauze



Suction filtration



Test tube



Watch glass

Crucible tongs

Safety Guidelines

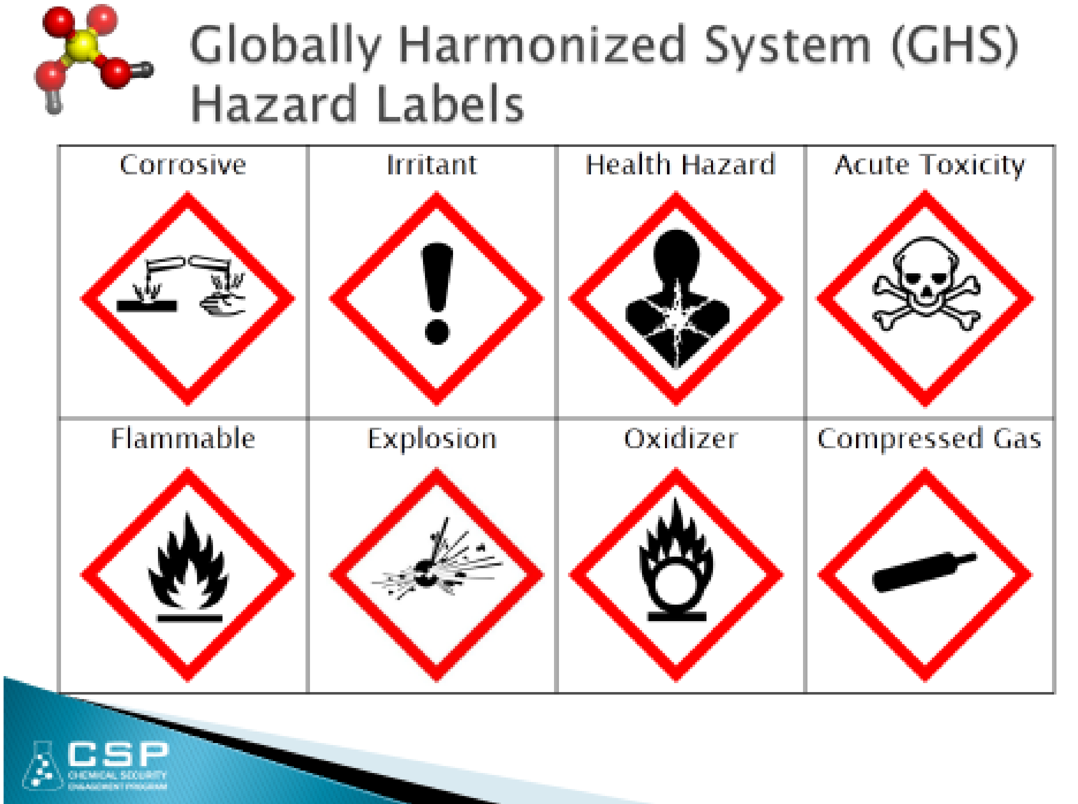
Life threatening injuries can happen in the laboratory. For that reason, students need to be informed of the correct way to act and things to do in the laboratory. The following is a safety checklist that can be used as a handout to students to acquaint them with the safety do and don't in the laboratory:

1. Wear your lab coats before you enter the lab area.
2. Keep the bench top clean, do not stack empty and unused bottles on your working area
3. Disinfect your working area with the proper disinfectant
4. Learn the location of fire extinguishers and how to use them.
5. Know the location of the emergency phone and doors.
6. Keep your laboratory workspace and equipment clean before and at the end of the laboratory session.
7. Wear approved eye protection (i.e., goggles).
8. Notify your instructor immediately in case of an accident.
9. Use disinfectant gel (HiGeen) before and after the experiment.
10. Wear disposable gloves when handling hazardous materials. Remove the gloves before exiting the laboratory.
11. Use the laboratory chemical hood, when there is a possibility of release of any unpleasant odor, toxic chemical vapors, dust, or gases. When using a hood, the sash opening should be kept at a minimum to protect the user and to ensure efficient operation of the hood. Keep your head and body outside of the hood face. Chemicals and equipment should be placed at least six inches within the hood to ensure proper airflow.
12. Wear shoes that adequately cover the whole foot; (no sandals or flip-flops).
13. Don't wear short trousers or skirts.
14. Bags, and other personal items must be stored in designated areas, not on the bench tops.
15. Don't wear contact lenses in laboratory.
16. Remove jewelry (especially dangling jewelry).
17. Synthetic finger nails are not recommended in the laboratory; they are made of extremely flammable polymers which can burn to completion and are not easily extinguished.
18. Use a hot water bath to heat flammable liquids. Never heat directly with a flame.

19. Keep your hands away from your face, eyes, mouth, and body while using chemicals.
20. Don't taste or sniff chemicals.
21. Check the label to verify it is the correct substance before using it.
22. Don't directly smell any vapor or gas, but fan it towards your nose.
23. Confine long hair.
24. Don't eat, drink, and smoke in laboratories.
25. Consider all chemicals to be hazardous unless you are instructed otherwise.
26. Dispose of chemicals as instructed by your instructor.
27. Dispose any broken glass only in the marked containers, not in the waste-baskets and dispose items contaminated with body fluids in the specified yellow bag.
28. If chemicals come in contact with your skin or eyes wash with copious amounts of water or eye wash and then consult your instructor.
29. Never point a test tube that you heat towards your face or your neighbor because it may erupt.
30. Do not perform any unauthorized experiment.
31. Always Add concentrated acid to water slowly. Never add water to a concentrated acid, because the heat of solution will cause the water to boil and the acid will spatter.
32. Never mix or use chemicals not called for in the laboratory exercise.
33. Avoid rubbing your eyes unless you know that your hands are clean.
34. Don't pour chemical waste into the sink drains or wastebaskets.
35. Place chemical waste in appropriately labeled waste containers.
36. Properly dispose of sharp objects (e.g., syringe needles) immediately in designated containers.
37. Many common reagents, like ether, alcohols, and acetone are highly flammable. Do not use them anywhere near open flame.

Reference:

School Chemistry Laboratory Safety Guide. Department of health and human services Centers for Disease Control and Prevention National Institute for Occupational Safety and Health.


Globally Harmonized System (GHS) Hazard Labels
Corrosive
Irritant
Health Hazard
Acute Toxicity
Flammable
Explosion
Oxidizer
Compressed Gas
CSP
CHEMICAL SECURITY ENGAGEMENT PROGRAM
1


Globally Harmonized System (GHS) and Other Hazard Labels
Environmental
Electricity
Hot Surface
Pinch Point
Biohazard
Radioactive
Optical Hazard
Laser
CSP
CHEMICAL SECURITY ENGAGEMENT PROGRAM
3

Experiment number 1: Density

1-Significant Figures (S.F):

1-الصفر بدون فواصل لا ينحسب كرقم معنوي سواء كان على اليمين او اليسار

2-الصفر بين اي رقمين معنوي

1. Counting of S.F ex: 1, 2, 3, 4, 5, 6, 7, 8, 9.
2. Zero in beginning of number isn’t S.F ex: 0089 2 S.F
3. Zero between two number is S.F ex: 2019 4 S.F
4. Zero in the end number is S.F there is decimal point
   Ex: 0.0050460 5 S.F and 700.0 3 S.F and 985 S.F and 400 1 S.F

1-Addition and subtraction and division and multiplication:

في حالة الجمع والطرح نأخذ اقل عدد عشري بعد الفاصلة العشرية

Ex: Find answer in lowest decimal point:

1) 7.1+2.25= 9.35 ~ 9.4 2 S.F round off 5 Inc. and less 5 not change.

2) 9.22-3 = 5.22 ~ 5 1 S.F

3) 8.23/2.1= 3.919 ~ 3.9 2 S.F

4) 3*2.2= 6.6 ~ 7 1 S.F

في حالة القسمة والضرب نأخذ العدد كامل

قاعدة التدوير ننظر الى اخر رقم نريد حذفه, اذا كان 5 او اكبر نحذفه ونزيد 1+ للرقم الذي قبله

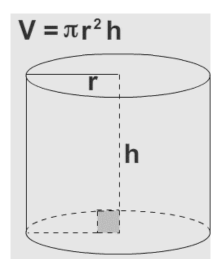
Density (D): D=Mass/Volume (gram/mL or Cm^3) SI Unit (Kg/m^3)

1 ml = 1 Cm^3 =1 CC

Loading balance mass and volume: (الميزان والمخبار المدرج)

Mass: Ex: 3.3 ± 0.1 or 3.33 ± 0.01 or 3.333 ± 0.001

Volume: Use Cylinder: 0.1 ~ 0.1 / 2 =0.05 or 0.2 ~ 0.2/2 = 0.1



Solid two type:

1-Regular 3D: cylinder: $V= \pi r^2 h$

Cubic: $V= L^3$

2-Irregular (Use Archimedes) ~ Concave (high surface tension)

Density depend two factors:

1) Purity 2) temperature as Increase volume increase and density decrease.

Physical property:

1) Intensive independent amount of substance ex: Density, Temperature, and Vapor.

2) Extensive depend on amount of substance ex: mass, heat, volume, energy.

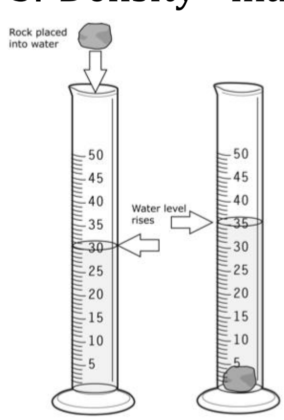
Accuracy: how close a measured value is to the **"true"** value being measured.

Precise: repeated measurements of the same subject/situation are very similar.

In Lab:

Part A: density of solid:

1. Mass of solid…………g ±……….
2. Volume of water……… Cm^3±………
3. Volume of water + metal……… Cm^3
4. Volume of metal (step 3 – step 2)……….Cm^3
5. Density =mass/volume (step 1 / step 4)………g/Cm^3



Part B: density of liquid:

1. Mass of cylinder empty...........g ±.........
2. Volume of water (H_2O) .... Cm^3±.........(Distil water)
3. Mass of water + cylinder...........g±.........
4. Mass of water = (step 3- step 1)g
5. Density =Mass / Volume (step 4/step 2)..........g/ Cm^3

Questions:

1) A solid ball has a mass of 50 grams and a volume of 20 cm^3. What is the density?
D=Mass/Volume
D=50/20= 2.5 g/cm^3

2) A solid cylinder has a radius of 2 cm and a length of 7 cm. It has a density of 3.1 g/cm^3. What is the mass of the cylinder?
Volume of cylinder= (π x r^2 x L) = (3.14 x 4 x 7) = 87.92 cm^3
D=Mass/Volume......... Mass= d x v= 3.1g/$\cancel{cm^3}$ x 87.92 $\cancel{cm^3}$=272.6 g.

3) The volume water of cylinder 7.7cm^3, after added of small piece iron metal (2.5 g) change of volume to 9.8 cm^3, calculate the density?
Volume= 9.8-7.7=2.1 cm^3
D= Mass/Vol.= 2.5/2.1= 1.19~ 1.2 g/cm^3 (don't forget S.F)

4) Which one of the following numbers contains 4 significant figures?

A)0.002960 B) 0.0029045 C) 0.00028 D) 0.00200650
E) 0.00570 Answer: **A**

5) If the density of Hexane is 0.943 g/cm^3 at 25 ^{0}C the volume (m^3) at 4.76 g of Hexane is (use S.F)?

A)5.05 X 10^{-6} B) 5.0 X 10^{-6} C) 4.40 X 10^{-6} D) 5.047 X 10^{-6} Answer:**A**

$$D = \frac{mass}{volume}$$

$$0.943 = \frac{4.76}{volume}$$

$$Volume = \frac{4.76}{0.943}$$

Volume= 5.0477 round off 5.05

Volume= 5.05 X 10^{-6}m

6) In which one of the following weights is the precision of balance is ± 0.001?

A)4.3498 g   B) 4.369 g   C) 4.89 g   D) 8.3 g   Answer: **B**

7) The mass of 8.9 ml of water was measured to be 7.98 g at 25 ^{0}C, the density of water (g/ml) is (the answer should be in correct S.F)?

A)0.99   B) 1.0   C) 0.90   D) 0.896   Answer: **C**

D = mass/volume → D = 7.98g/8.9ml → D = 0.896 g/ml → D = 0.90 g/ml.

8) If the density of Toluene is 0.589 g/mL at 25 ^{0}C the volume (L) at 8.69 g of Toluene is (use S.F)?

9) The volume water of cylinder 9.5 mL, after added of small piece Cupper metal (1.5 g) change of volume to 14.5 mL, calculate the density?

10) A solid ball has a mass of 40 grams and a volume of 17.6 cm3. What is the density?

Experiment number 2: Physical Method for Separation of Mixture

Mixture: inert components in certain contains.

Two type of mixtures:

1. Homogenous components have same chemical and physical properties ex: (air, solution $NaCl/H_2O$)

2. Heterogeneous differ in chemical and physical properties ex: (sand/oil, $NaCl_{(s)}/SiO_{2(s)}/NH4Cl_{(s)}$).

التسامي: تغير من الحالة الصلبة الى الغازية بدون المرور بالحالة السائلة.

Properties of **unknowns**: ($NaCl$, SiO_2, NH_4Cl):

نقوم بالتخلص من NH_4Cl

Melting point: $SiO_2 > NaCl > NH_4Cl$ in solid state.

Heating of mixture in Bunsen burner: observe evolving of white fumes $NH4Cl_{(s)} \xrightarrow{Heating} NH_4Cl_{(g)}$ (Sublimation) all NH_4Cl is removed.

Sublimation: is the physical property of some substances to pass directly from the solid state to the gaseous state without the appearance of the liquid state.

Solubility in water: $NaCl$, NH_4Cl ~~ Soluble / /SiO_2 ~~ insoluble.

Add distilled water to remained mixture to dissolve NaCl

$NaCl \xrightarrow{dissolving} Na^+ + Cl^-$ (Extraction) and filtration of SiO_2.

الاستخلاص: هي عملية نستخدمها للفصل بين المكونات عن طريق الذوبان (يذوب او لا يذوب)

Extraction: is the removal of one substance from a mixture because of its greater solubility in a given solvent.

الفلترة: عملية فصل بين مادة صلبة و مادة سائلة بأستخدام ورقة الترشيح

Filtration: is the process of removing or "straining" a solid (the chemical term is precipitate) from a liquid by the use of filter paper or other porous material

In Lab:

1. Mass of empty evaporating dish ………. g
2. Mass of original Mixture 2 g and evaporating dish……..g (وزن وهو على الميزان)
3. Mass of dish and remained mixture ………g ($NaCl + SiO_2$)
4. Mass of NH_4Cl = mass of dish and mixture before heating(step2) - ~~after heating (step3)

بعد التسخين سنفقد وزن ال NH_4Cl

5. Mass one/two filter paper (F.P)…………g
6. Mass of SiO_2 in Mixture = mass of SiO_2 and F.P after drying – mass of filter paper.

بعد عملية الفلترة سيبقى لدينا وزن ال SiO_2

7. Mass of NaCl = mass of original mixture before heating – (mass NH_4Cl + mass of SiO_2).

بعد التجفيف أصبح لدينا وزن $NH_4Cl + SiO_2$

8. Practically: Vaporization of filtration Solution:
   % of each component in original mixture = mass of components / mass of original mixture (2 g) X 100%

نقوم بحساب وزن NaCl

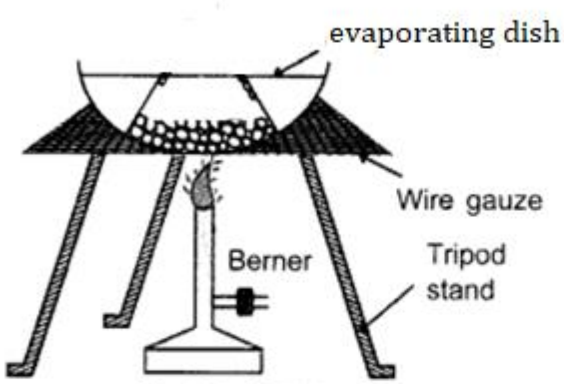
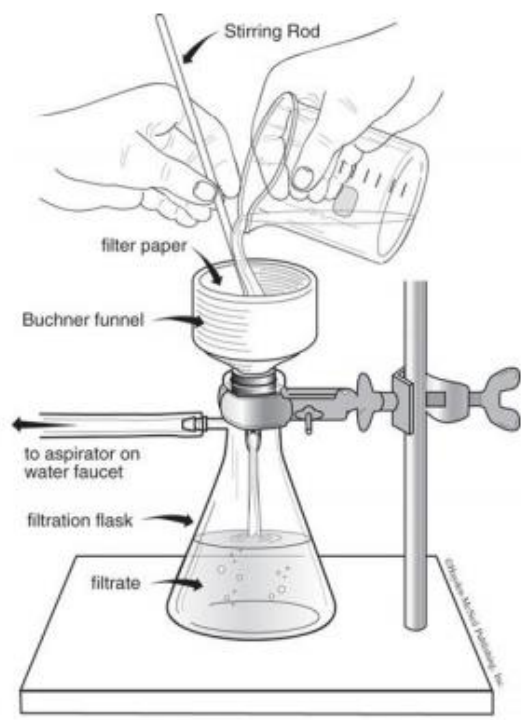


Fig. 7.1. Sublimation



Separation Technique: Suction Filtration

Questions:

1) Which of the following statements are true?

1) SiO_2 is insoluble in water
2) The melting point of NH_4Cl is higher than that of SiO_2
3) NH_4Cl is soluble in water but NaCl is insoluble
4) SiO_2 could be separated from NaCl solution by filtration

A)2,3 B) 1,4 C) 1,2 D)2,4 Answer: **B**

2) A mixture weighing 9.35 g containing (NaCl, NH_4Cl and 1.2 g of SiO_2) when this mixture was heated its mass was decreased to 7.15 g, the mass % of NaCl in the mixture is:

A) 63.6% B) 53% C) 42.2% D) 31.5% Answer: **A**

9.35 g = mass NH_4Cl + mass SiO_2+ mass NaCl

9.35 g = mass NH_4Cl + 1.2 g+ mass NaCl

After heating $\rightarrow$ 9.35 – 7.15 = 2.2 g (mass of NH_4Cl)

9.35 g = 2.2 g+ 1.2 g + mass NaCl

Mass NaCl = 9.35 g -2.2 g- 1.2 g $\rightarrow$ mass NaCl = 5.95 g

Mass NaCl%= 5.95/9.35 $\rightarrow$ Mass NaCl% = 63.6%

3) A mixture containing (0.5 g SiO_2, 35% (wt/wt) (NaCl)) and unknown amount of NH_4Cl when the mixture was heated its mass was decreased to 1.3 g the mass of mixture is:

A) 5.7 g B) 7 g C) 1.5 g D) 2.28 g Answer: **D**

Mass after heating= mass SiO_2+ mass NaCl

1.3= 0.5 + X $\rightarrow$ X=1.3-0.5= 0.8$\rightarrow$ Mass of NaCl= 0.8 g

35% of NaCl:

$$\text{NaCl \%} = \frac{\text{Mass NaCl}}{\text{Total mass}} \times 100\%$$

$$\frac{35\ \%}{100\ \%} = \frac{0.8}{\text{Total mass}} \times \frac{100\%}{100\%}$$

$$0.35 = \frac{0.8}{\text{Total mass}}$$

$$\text{Total mass} = \frac{0.8}{0.35} = 2.28$$

4) Continue with Question 3; Calculate mass NH_4Cl?

A)0.7 B) 2.0 C) 1.2 D) 0.98 Answer: **D**

Mass of sample= Mass NH_4Cl + Mass SiO_2 + Mass NaCl

2.28 = Mass NH_4Cl + 0.5 + 0.8

Mass NH_4Cl= 2.28 – (0.8 + 0.5) = 0.98 g

5) Which of the following Glassware is used in lab for sublimation technique:

A)Watch glass B) Evaporating dish C) Graduated cylinder D) Buchner funnel Answer: **B**

Experiment number 3: Limiting Reactant

Limiting Reactant is the reactant that consumed completely and the first and limit or determine the amount of product.

Example: $3A + 2B \longrightarrow 4C + 3D$ when 6 moles of A react with 5 moles of B:

1. Mole ratio of A = 6/3 = 2 is the limiting reactant.
2. Mole ratio of B = 5/2 = 2.5 is the excess.

الخطوة الاولى والثانية فقط لتحديد
العامل المحدد. **لا تدخل** في الحسابات

3. How many moles of A reacted = 6
4. How many moles of A remained = 0
5. How many moles of B reacted = calculate?

   3 mole of A ——— 2 mole of B

   6 ——— X

   X= 6x2/3 = 4 moles of B reacted

دائما" لتحديد عدد مولات شيء محدد
الشيء الذي تسأل عنه تكون **معاملته في البسط**

في اي عملية حسابية يكون العامل
المحدد هو صاحب القرار

6. How many moles of B remained = initial – reacted = 5-4 =1.
7. How many moles of C?

   Mole of C = 4/3 mole of A= 4/3 X 6= 8.

لاحظ عندما سألت عن مولات C
كان معاملها في البسط

Unknown's: $BaCl_2.2H_2O + Na_3PO_4.12H_2O$

Equations: $3BaCl_2.2H_2O + 2Na_3PO_4.12H_2O \longrightarrow Ba_3(PO_4)_{2\ P.P.t} + 6NaCl + 30\ H_2O$

$3Ba^{2+} + \cancel{6Cl^-} + \cancel{6H_2O} + \cancel{6\ Na^+} + 2PO_4^{3-} + \cancel{24\ H_2O} \longrightarrow Ba_3(PO_4)_{2\ P.P.t} + \cancel{6Na^+} + \cancel{6Cl^-}$

Ionic equation: $3Ba^{2+} + 2PO_4^{3-} \longrightarrow Ba_3(PO_4)_{2\ P.P.t}$

Na^+, Cl^-: spectator ions.

In Lab:

1 g of unknown's added to 200 ml D.W in water path 30 min at 80 °C.

- after 30 min in water path remove beaker, in beaker liquid two layers (supernatant +contains of excess unreacted Ba^{2+} or PO_4^{3-} and $Ba_3(PO_4)_{2\ P.P.t}$)
- Determent of limiting reactant: decantation of supernatant

لتحديد العامل المحدد عمليا" نقوم بسحب كمية من الرائق الى [2 bukers]

ثم نضيف:

Add (0.5M) $Na_3PO_4.12H_2$ and

Add (0.5M) $BaCl_2.2H_2O$

Add (0.5M) $Na_3PO_4.12H_2O$

Add (0.5M) $BaCl_2.2H_2O$

المحلول الذي يتعكر بعد الاضافة هو العامل المحدد

Beaker Sour (تعكر) solution is limiting reactant.

- Drying of $Ba_3(PO_4)_2$ for 30 min in oven.
- Weighing of $Ba_3(PO_4)_2$ ……….. g.

Calculation:

1. Mass of $Ba_3(PO_4)_2$ = mass of p.p.t and filter paper – mass filter paper
2. Moles of $Ba_3(PO_4)_2$ = mass / molar mass = step 1/602.2
3. Moles of $BaCl_2.2H_2O$ reacted complete = moles of $Ba_3(PO_4)_2$ X 3/1
4. Mass of $BaCl_2.2H_2O$ reacted complete = step 3 X 244.2
5. Moles of $Na_3PO_4.12H_2O$ reacted complete = moles of $Ba_3(PO_4)_2$ X 2/1
6. Mass of $Na_3PO_4.12H_2O$ reacted complete = step 5 X 380.2
7. Mass of excess reactant (react + unreacted) = mass of mixture (1g) – mass of limiting reactant

8. Mass % of $BaCl_2$ = mass of $BaCl_2$ / 1g = (step 4 L.R or step 7 excess)

9. Mass % of Na_3PO_4 = mass of Na_3PO_4 / 1g = (step 6 L.R or step 7 excess) Total mass 100%

Questions:

1) Why use water path at 70-80 °C?

We don't want to get to the boiling point.

2) Filtrate of the $Ba_3(PO_4)_2$ wash of precipitate with water or basic medium no wash with acidic medium? why?

Washing with acidic medium will reduce (decrease) actual yield.

3) mixture weighing 3.55 g containing $Na_3PO_4.12H_2O$ [M.M=380.2 g/mol] and $BaCl_2.2H_2O$ [M.M= 244.2 g/mol] when this mixture was dissolved around 200 ml distilled water and warm with water-path for 30 minutes. A 0.452 g of $Ba_3(PO_4)_2$ [M.M= 602.2 g/mol] was formed according to the following of ionic equation:

$3Ba^{2+} + 2PO_4^{3-} \longrightarrow Ba_3(PO_4)_{2\ P.P.t}$; if $Na_3PO_4.12H_2O$ is limiting reactant.

Calculate:

I. How many grams of limiting reactant ($Na_3PO_4.12H_2O$)?
II. How many grams excess (unreacted)?
III. % of each $Na_3PO_4.12H_2O$ and $BaCl_2.2H_2O$?

i-Moles of $Ba_3(PO_4)_2 = \frac{Mass}{M.M} = \frac{0.452}{602.2} = 7.5X10^{-4}$ mole.

Moles of $Na_3PO_4.12H_2O = \frac{2}{1}$ X $7.5X10^{-4} = 1.5X10^{-3}$ mole.

Mass of $Na_3PO_4.12H_2O$ reacted complete = $1.5X10^{-3}$ X 380.2= 0.5703 g.

ii-Moles of $BaCl_2.2H_2O = \frac{3}{1}$ X $7X10^{-4}$ = $2.25X10^{-3}$ mole.

Mass of $BaCl_2.2H_2O$ reacted complete = $2.25X10^{-3}$ X 244.2= 0.55 g.

Mass of excess (react + unreacted) = mass of mixture - mass of L.R

Mass of excess (react + unreacted) = 3.55 g - 0.5703= 2.98 g.

Mass of excess (react + unreacted) = 2.98 g

Mass of excess (0.55+ unreacted) = 2.98 g

Mass of excess (unreacted) = 2.98 - 0.55= 2.4297g

iii-Mass % of $BaCl_2.2H_2O = \frac{2.98}{3.55}$ X 100% = 83.9 %

Mass % of $Na_3PO_4.12H_2O= \frac{0.57}{3.55}$ X 100% = 16.1 %

4) A mixture weighing 5.44 g containing only $BaCl_2.2H_2O$ (M.wt = 244.2 g/mol) and $Na_3PO_4.12H_2O$ (M.wt = 380.2 g/mol) when this mixture was dissolved in 200 ml distilled water and heated for 30 minutes , A 0.765 g of $Ba_3(PO_4)_2$ (M.wt = 602.2 g/mol) was formed according to the following equation:

$$3Ba^{+2} + 2\ PO_4^{-3} \longrightarrow Ba_3(PO_4)_2$$

If $BaCl_2$ is limiting reactant in this mixture, how many grams of $Na_3PO_4.12H_2O$ unreacted in this mixture:

A) 1.68 g B) 3.54 g C) 3.68 g D) 4.68 g Answer: **B**

Mole of $Ba_3(PO_4)_2$ = mass/M.M $\longrightarrow$ 0.765/602.2

Mole of $Ba_3(PO_4)_2$ = 1.27 X 10^{-3}

3 mole $Ba^{+2} \longrightarrow$ 1 mole $Ba_3(PO_4)_2$

????????? ? $\longrightarrow$ 1.27 x 10^{-3} mole

Mole Ba^{+2} = 3.81 x 10^{-3}

Mass of Ba^{+2} = mole Ba^{+2} X M.M

Mass of Ba^{+2} = 3.81 X 10^{-3} X 244.2

Mass of Ba^{+2} = 0.931 g

3 mole $Ba^{+2} \longrightarrow$ 2 mole PO_4^{-3}

3.81 X 10^{-3} $\longrightarrow$??????????

Mole PO_4^{-3} = 2.54 X 10^{-3}

Mass of PO_4^{-3} = mole PO_4^{-3} X M.M

Mass of PO_4^{-3} = 2.54 X 10^{-3} X 380.2

Mass of PO_4^{-3} = 0.966 g

Mass of $Na_3PO_4.12H_2O$ unreacted = mass of mixture - mass of limiting reactant- mass of PO_4^{-3} reactant:

Mass of $Na_3PO_4.12H_2O$ unreacted = 5.44 g - 0.931 g - 0.966 g

Mass of $Na_3PO_4.12H_2O$ unreacted = 3.54 g

5) In the limiting reactant experiment, if $Na_3PO_4.12H_2O$ solution added to supernatant and white precipitate is formed it means that:

A) $BaCl_2.2H_2O$ is limiting reactant and $Na_3PO_4.12H_2O$ is excess reactant.
B) $Na_3PO_4.12H_2O$ is limiting reactant and $BaCl_2.2H_2O$ is excess reactant.
C) Both are limiting reactant.
D) Both are excess reactant.
Answer: **B**

6) If 0.8 moles $BaCl_2.2H_2O$ (M.wt = 244.2 g/mol) and mixed with 0.9 mol of $Na_3PO_4.12H_2O$ (M.wt = 380.2 g/mol). How many grams of $Ba_3(PO_4)_2$ (M.wt = 602.2 g/mol) will be produced:

$$3BaCl_2.2H_2O + 2Na_3PO_4.12H_2O \longrightarrow Ba_3(PO_4)_2 + 6NaCl + 30H_2O$$

A)140.65 g B) 160.58 g C) 240.9 g D) 180.7 g Answer: **B**

Solution:

Mole ratio of $BaCL_2.2H_2O$ = 0.8/3 = 0.267 $\longrightarrow$ Limiting reactant
Mole ratio of $Na_3PO_4.12H_2O$ = 0.9/2 = 0.45 $\longrightarrow$ Excess reactant

3 mol $BaCl_2.2H_2O$ $\longrightarrow$ 1 mol $Ba_3(PO_4)_2$
0.8 mol $\longrightarrow$???????????

mol $Ba_3(PO_4)_2$ = 0.267

Mole = mass/M.M $\longrightarrow$ Mass = mole X M.M
Mass = 0.267 X 602.2
Mass = 160.58 g

7) A mixture of $BaCl_2.2H_2O$ (M.wt = 244.2 g/mol) and $Na_3PO_4.12H_2O$ (M.wt = 380.2 g/mol) weighing 6.50 g was dissolved in distilled water if 2.3 g $Ba_3(PO_4)_2$ (M.wt = 602.2 g/mol) was precipitated according to:

$$3BaCl_2.2H_2O + 2Na_3PO_4.12H_2O \longrightarrow Ba_3(PO_4)_2 + 6NaCl + 30H_2O$$

If $Na_3PO_4.12H_2O$ is the limiting reactant, then the mass percent of the excess reactant in the mixture is:

A)64% B) 57.3% C) 75.2% D) 60.0% E) 54.5%

Answer: **E**

Mole $Ba_3(PO_4)_2$ = mass/ M.wt $\longrightarrow$ Mole = 2.3/602.2 = 3.82 X 10^{-3}

2 mol $Na_3PO_{4.}12H_2O \longrightarrow$ 1 mol $Ba_3(PO_4)_2$

??????????????? $\longrightarrow$ 3.82 X 10^{-3}

Mole $Na_3PO_{4.}12H_2O$ = 7.64 X 10^{-3}

Mass of$Na_3PO_{4.}12H_2O$ = mol X M.wt

Mass of $Na_3PO_{4.}12H_2O$ = 7.64 X 10^{-3} X 380.2

Mass of $Na_3PO_{4.}12H_2O$ = 2.9 g

Mass of excess reactant = mass of mixture - mass of L.R
= 6.5 - 2.9
= 3.6 g

Mass $BaCl_{2.}2H_2O$ % = (3.6/6.5) X 100%
Mass $BaCl_{2.}2H_2O$ % = 55.3 %

8) which of the following statements are false?

1. Density is independent on the amount of substance.
2. SiO_2 is soluble in water while NH_4Cl is insoluble.
3. The melting point of NH_4Cl is lower than the melting point of NaCl.
4. $Ba_3(PO)_4$ is soluble in water.

A)1, 4 B) 1, 3 C) 2, 4 D) 3, 4 Answer: **C**

Experiment number 4: Chemical in Everyday Life

1-Glass Cleaner (NH_3)$_{aq}$ Ammonia active chemical species.

Observation litmus paper: **Red** to **Blue** NH_3 is base.

$NH_3 + H_2O \longrightarrow NH_4OH$.

2-Fertilizer (NH_4^+ Ammonium ion)

Small Sample of fertilizer (NH_4^+ acid) + NaOH or KOH $\longrightarrow$ $NH_{3(g)} + H_2O$

$NH_{3\,(g)}$: 1-Colorless 2-Base 3-Pungent odor (bad)

Observation litmus paper: **Red** to **Blue.**

3-Table Salt (NaCl) Cl^- active species

$NaCl_{(s)} \xrightarrow{\text{St. acid H+ } H_2SO_4 \text{ (18) molar}} Na^+ + HCl_{(g)}$

$HCl_{(g)}$: 1-Colorless 2- acid 3-pungent odor.

Observation litmus paper: **Blue** to **Red.**

Another Eq: NaCl + Silver nitrate ($AgNO_3$) $\longrightarrow$ $AgCl_{(s)} + NaNO_{3\,(aq)}$

$AgNO_3$ Conserved in brown bottle why? To prevent reduction of Ag^+ to Ag

4-Backing soda ($NaHCO_3$) $_{(s)}$:

$NaHCO_3$ + acid (HNO_3 or Vinegar) $\longrightarrow$ $Na^+ + H_2O + CO_2$ (g)

$CO_{2(g)}$: 1-colorless 2-oderless 3- acidic in water.

Passing of CO_2 to saturated solution $Ba(OH)_2$ colorless.

$CO_2 + Ba(OH)_2 \longrightarrow BaCO_{3(s)} + H_2O$, $BaCO_3$ white solid.

5-Bathroom-Cleaner (HCl) $_{(aq)}$: (flash)

Carbonate (solid) eg: $CaCO_3$ or Na_2CO_3 $\longrightarrow$ $CO_{2\,(g)} + CaCl_2 + H_2O$

$CO_{2(g)}$: 1-colorless 2-Odorless.

6-Bleach ($NaOCl)_{(L)}$: Sodium hypo chlorite

Solid Bleach: $Ca(OCl)_2$ Calcium hypo chlorite

Eq: $Cl_2 + OH^- \longrightarrow OCl^- + HCL +$ Bleach

General Eq: Iodide solution (KI) + Bleach $\longrightarrow$ Red Color $I_2(aq) \xrightarrow{\text{Starch}}$ Dark Blue Color (I_2.Strach)

Reduction

$$Cl_2 \text{ (green yellow)} + 2I^- \longrightarrow 2Cl^-\text{(red)} + I_2 \text{ (Redox Rxn)}$$

Oxidation

7-Drain-Opener (NaOH(s)/AL(s)):

Eq: $2NaOH_{(s)} + 2Al = 6H_2O \longrightarrow 2NaAl(OH)_4 + 3H_{2(g)}$

$H_{2\ (gas)}$: 1-high pressure 2-press the dirties.

Observation litmus paper: **Red** to **Blue.**

Questions:

1) CO_2 gas could be evolved through the addition of:

A) Drain opener to hot water. B) strong base to fertilizer.
C) Bathroom cleaner to table salt. D) Bathroom cleaner to limestone.

Answer: **D**

2) The active chemical ingredient in glass cleaner is:

A) $NaHCO_3$ B) HCl C) NH_3 D) NaCl Answer: **C**

3) NH_3 gas could be described as:

A) Colorless, Odorless and acid. B) Colorless, pungent odor and acid.

C) Colorless, pungent odor and base. D) Colorless, Odorless and base.

Answer: **C**

4) Which of the following statement is true?

I. In the reaction of Bleach solution with KI solution, I^- is the oxidizing agent.
II. Glass cleaner is diluted solution of Ammonia.
III. The litmus paper changes from blue to red in drain paper solution.
IV. Density decrease with increasing of temperature.

A)I, IV B) II, III C) I, III D) III, IV E) II, IV

Answer: **E**

5) HCl gas could be described as:

A) Colorless, odorless, acidic in water.
B) Colorless, pungent odor, acidic in water.
C) Colorless, pungent odor, basic in water.
D) Violet vapor, pungent odor.

Answer: **B**

6) H_2 gas could be evolved through the addition of:

A) Strong base to table salt. B) Strong base to fertilizer.
C) Strong acid to backing soda. D) Hot water to drain opener.

Answer: **D**

7) Which of the following is bathroom cleaner?

A) $NH_{3(aq)}$ B) $NaHCO_{3(s)}$ C) $HCl_{(aq)}$ D) $NaCl_{(s)}$ Answer: **C**

8) Which of the following reagents could from a white precipitate with NaCl Solution:

A) $Ba(OH)_{2(aq)}$ B) $BaCl_{2(aq)}$ C) $Starch_{(aq)}$ D) $AgNO_{3(aq)}$ Answer: **D**

9) which one of the following instruments is used in the lab to get rid of pungent odor?

A) Oven B) Balance C) fume – Hood D) fire extinguisher

Answer: **C**

10) The reagent (X) in the following equation is:

$$2NaCl_{(s)} + H_2SO_{4(aq)} \longrightarrow 2X_{(g)} + Na_2SO_{4(aq)}$$

A) HCl B) Cl_2 C) Cl^- D) NH_3 Answer: **A**

11) Which of the following reagents could be used to detect the presence of CO_2:

A) $AgNO_3$ B) HCl C) $Ba(OH)_{2\,(aq)}$ D) $HNO_{3\,(aq)}$ Answer: **C**

12) Which of the following Bath room cleaner:

A) $HCl_{(aq)}$ B) $CaCO_{3(s)}$ C) $NaCl_{(s)}$ D) $NaHCO_{3(s)}$ Answer: **A**

13) Addition of strong acid to table salt produces:

A) $CO_{2(g)}$ B) $NH_{3(g)}$ C) $AgCl_{(s)}$ D) $H_{2(g)}$ E) $HCl_{(g)}$

Answer: **E**

14) Ammonia gas produced by the reaction of:

A) Drain opener with hot water B) Strong acid with backing soda

C) Strong acid with table salt D) Fertilizer with strong base

Answer: **D**

15) Drain opener is:

A) NH_4^+ B) $NaOH_{(s)}$ /$Al_{(s)}$ C) $NH_{3(aq)}$ D) $CaCO_{3(aq)}$

Answer: **B**

16) Which of the following gases is colorless, acid and odorless:

A) $NH_{3(g)}$ B) $Cl_{2(g)}$ C) $HCl_{(g)}$ D) $CO_{2(g)}$

Answer: **D**

Experiment number 5: Colligative Properties.

Physical properties depend only on the amount (or concentration) of solute particles and not on <u>the nature or identity of solute</u> regarding the solute: volatile / not volatile / ionic / molecule.

* Important example:

1. Lowering in vapor pressure of solution.
2. Increase or elevation in boiling point.
3. Depression or decrease on freezing point.
4. Increasing in osmotic pressure.

 * **Osmosis:** movement of solvent molecules from lower concentration to higher concentration medium during semi-permeable membrane.

$\pi \;\alpha\; M$

$\pi_1 < \pi_2$

$M_1 < M_2$, M_1: hypotonic solution M_2: hypertonic solution.

Follow the freezing point of the bottom of test tube record T_f (solution) < 0 equilibrium constant temp.

$\Delta T_f = T_f$ Solvent – T_f Solution = 0 – (-ve) = +ve

$\Delta T_f \;\alpha\; m$………………m: molality of solution

$\Delta T_f = i \times K_f \times m$

i: no, of particles of solute.

i: no. of ion's for ionic compounds three type

1- Strong electrolyte i= no. of ion's

Ex: NaCl=2, $CaCl_2$= 3, $AlCl_3$= 4

2- Weak electrolyte = 1.5~2
3- Non electrolyte =1 ex: glucose, glycerol, and glycol….etc.

In lab:

* Depression in freezing point:

1. 30 ml(30g) of pure water ~ Density
2. T_f (water) = 0 °C (freezing point)
3. Add 2.5 g of molecules solute.
4. Shaking for Homogenous solution.

Ice + NaCl? To dissolve ice to prevent super cooling to

Obtain smooth cooling (graduated).

Questions:

Q 1) What is the freezing point of a solution of 15.0 g of NaCl in 250 g of water? The molal freezing point constant, K_f, for water is 1.86 ^oC kg / mol.

Answer:

For NaCl, i = 2.

$$Mole_{solute} = \frac{15}{58.44} = 0.257$$

$$m = \frac{Mole_{solute}}{Kg_{solvent}} = \frac{0.257}{250 \text{ X } 10^{-3}} = 1.03$$

$\Delta T_f = i \text{ x } K_f \text{ x } m$

$\Delta T_f = 2 \text{ x } 1.86 \text{ x } 1.03$

$\Delta T_f = 3.83$, Therefore, the *change* in the freezing point of the water is -3.8 ^oC. The freezing point of the solution is, therefore, -3.8 ^oC.

Q2) Ethylene glycol (EG), $CH_2(OH)CH_2(OH)$, is a common automobile antifreeze. Calculate the freezing point of a solution containing 651 g of EG in 2505 g of water? The molar mass of EG is 62.01 g/mol. $K_f = 1.86$ $^oC/m$.

Answer:

$$Mole_{solute} = \frac{651}{62.1} = 10.5$$

$$m = \frac{Mole_{solute}}{Kg_{solvent}} = \frac{10.5}{2.505} = 4.19$$

$\Delta T = i \text{ x } K_f \text{ x } m = (1.86\ ^oC/m)(4.19\ m) = 7.79\ ^oC$

Since pure water freezes at 0 ^oC, the solution will freeze at -7.79 ^oC.

Q3) A 7.85 g sample of a compound with empirical formula C_5H_4 is dissolved in 301 g of benzene. The freezing point of the solution is 1.05 ^oC below that of the pure benzene. What is the molar mass? K_f for benzene is 5.12 $^oC/m$.

Answer:

$$m = \frac{\Delta T_f}{K_f} = \frac{1.05}{5.12} = 0.205$$

$$m = \frac{Mole_{solute}}{Kg_{solvent}} = \frac{Mole_{solute}}{301 \text{ X } 10^{-3}} = 0.205$$

$$Mole_{solute} = 301 \text{ X } 10^{-3} \text{ X } 0.205 = 0.0617 \text{ mol}$$

$$Mole_{solute} = \frac{Mass}{M.Mass}$$

$$0.0617 = \frac{7.85}{M.Mass}$$

$$M.Mass = \frac{7.85}{0.0617} = 127 \text{ g/mol}$$

Q4) As the amount of solute added to solution increase?

A) Freezing point will increase.
B) Boiling point will decrease.
C) Vapor pressure will decrease.
D) Osmatic pressure will decrease.

Answer: **A**

Q5) Which of the following statements is **FALSE** as the molality of solute increase:

A) Decrease the freezing point of a solution.
B) Increase the boiling point of a solution.
C) Increase the vapor pressure of a solution.
D) Increase the osmatic pressure of a solution.

Answer: **C**

Q6) Which of the following statements is (are) false:

I) Density is independent on temperature
II) Air is homogeneous mixture
III) Melting point of NH_4Cl is lower than that of SiO_2
IV) Increasing of Osmotic pressure is not a colligative property

A) I, IV B) II, III C) I, II D) I, III

Answer: **A**

Q7) The true of statements of the following are:

I. The freezing point of solution decrease with increasing the amount of solute.
II. The boiling point of solution decrease with increasing the amount of solute.
III. The molal freezing point constant K_F depends on the type of solvent.
IV. SiO_2 is soluble in water while NH_4Cl is insoluble.

A) I and III B) II and IV C) I and IV D) II and III

Answer: **A**

Q8) When 13.6 g of unknown solute was added to 150g of water (T_f = 0^0C, K_f = 1.86) the freezing point of the resulting solution was recorded to be -3^oC the molar mass of solute is:

A) 67.5 B) 84.3 C) 56.2 D) 112.4 Answer: **C**

$\Delta T_f = K_f \times \underline{m}$

$$0-(-3) = 1.86 \times \frac{\text{Mole of solute}}{\text{Mass of solvent Kg}}$$

$$0-(-3) = 1.86 \times \frac{\text{Mass of solute}}{\text{M.M} \times \text{Mass of solvent Kg}}$$

$$3 = 1.86 \times \frac{13.6}{\text{M.M} \times 0.15}$$

M.M = 56.2g/mol

Creativity

Q9) How many grams of glycerol (M.M = 92 g/mol) should be add to 200 g of water (T_f = 0^0C, Kf = 1.86) to decrease the freezing point by 3^0C below the freezing point of water:

A)24.7 g B) 19.8 g C) 34.6 g D) 29.7 g Answer: **D**

$\Delta T_f = K_f \text{ X } \underline{\underline{m}}$

$$0-(-3) = 1.86 \text{ X } \frac{\text{Mole of glycerol}}{\text{Mass of solvent Kg}}$$

$$0-(-3) = 1.86 \text{ X } \frac{\text{Mass of glycerol}}{M.M_{glycerol} \text{ X Mass of solvent Kg}}$$

$$3 = 1.86 \text{ X } \frac{\text{Mass}}{92 \text{ X } 0.2}$$

Mass = 29.7 g

Q10) Calculate the freezing point of a solution made by dissolving 30 g of ethylene glycol (M.M = 62 g/mol) in 70 g of water ($T_f = 0^0C$, K_f = 1.86 C/m)

A) 12.9^0 C B) -24.5^0 C C) 24.5^0C D) -12.9^0C Answer: **D**

$\Delta T_f = K_f \text{ X } \underline{\underline{m}}$

$\Delta T_f = T_{fsolvent} - T_{fsolution} = K_f \text{ X } m$

$$0 - T_{f(Sol'n)} = K_f \text{ X } \frac{\text{Mole of solute}}{\text{Mass of solvent Kg}}$$

$$0 - T_{f(Sol'n)} = 1.86 \text{ X } \frac{\text{Mass of ethylene glycol}}{M.M_{ethylene\ glycol} \text{ X Mass of solvent Kg}}$$

$$-T_{f(Sol'n)} = 1.86 \text{ X } \frac{30}{62 \text{ X } 0.07}$$

$-T_{f(Sol'n)} = 12.9$ g $\longrightarrow T_{f(Sol'n)} = -12.9$ g

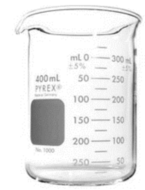
Final

Glassware

Graduated cylinder

Spatula

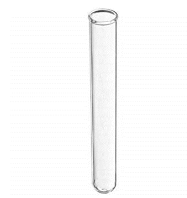
Bunsen burner



Beaker

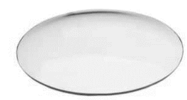
Evaporating dish

Glass rod

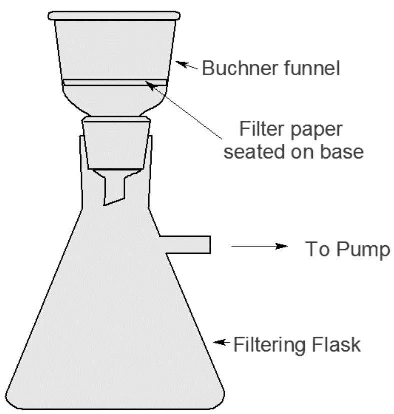


Test tube

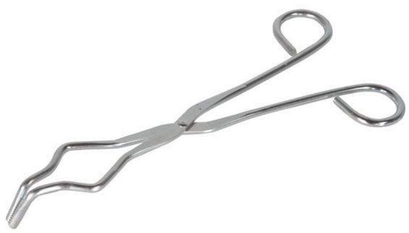
Wire gauze



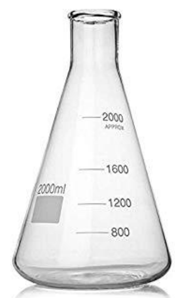
Watch glass



Suction filtration



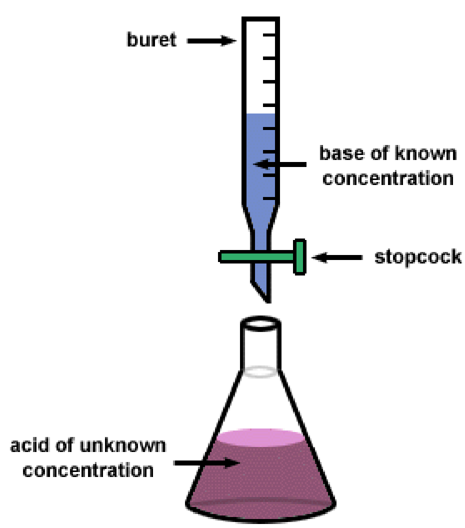
Crucible tongs



Erlenmeyer flask

pipette

Thermometer



Centrifuge

Experiment number 6: Calorimetry

Calorimetry: is advice or tool that could be used to measure of the amount and direction of the evolved (lost) and absorbed heat for any chemical and physical change.

جهاز يستحدم لقياس التغير في الحرارة سواء كانت مفقود او مكتسبة

Two type reaction: 1- Endothermic: heat °C is +ve

2- Exothermic: heat °C is -ve

Calorimetry two type:

هناك نوعان:

1-Colsed: Bomb-calorimetry (measure heat at constant volume

النوع الاول : المغلق

$q_v = \triangle E$ E: energy)

نقيس الحرارة بثباث الحجم

2-Simple or opened or coffee cup (polystyrene [good isolator]

النوع الثاني : المفتوح

نقيس الحرارة بثبات الضغط

To consume or keep the heat (measure the heat at constant pressure $q_p = \triangle H$ H: enthalpy)

-You follow increasing up or dawn (decrease).

In Lab:

Part A: Heat of solution (dissolving of NaOH in water)

Eq: NaOH (2g) + water (100 g or ml)

1. 100 ml of pure water record T_i
2. Add of NaOH (2g) and increase temperature (exothermic) until the temperature is constant.
3. Record T_f (T_{max})

Equation: $NaOH + H_2O \longrightarrow Na^+ + OH^- + \text{heat } \triangle H$

$\triangle$ H= - ($\triangle$T X mass of sol'n 102 g X specific heat 4.18J/g.°C)

$\triangle T = (T_f - T_i)$

Specific heat (**intensive property**): is heat required to raise the temperature one gram of substance one degree °C or K.

بدنا نحول الوحدة الى

KJ/mol

$\triangle H_{KJ/mol}$= - (4.18 X$\triangle$ T X 102g)/ (1000 X moles of NaOH (L.R))

قسمنا على مولات العامل المحدد

NaOH

Moles of NaOH= Mass/Molar Mass = 2/40=0.05 mole

وقسمنا على 1000

Part B: heat of neutralization 2g of NaOH + 100 ml HCl

1. 100 ml HCl and record T_i
2. Add 2g of NaOH and record T_f at max exothermic

$\triangle$T (part B) >>> $\triangle$T (part A) {double of change}

المعادلة الاولى مماثلة لمعادلة الجزء الاول

Part A: $NaOH_{(s)} + H_2O \longrightarrow Na^+ + OH^-$ + heat $\triangle H_1$

Part B: $NaOH_{(s)} + HCl/ H_2O \longrightarrow Na^+ + OH^-$ + heat $\triangle H_1$

المعادلة الثانية يظهر نفس التفكك ولكن يظهر جزء ال

Neutralizer

$OH^- + H^+ \xrightarrow{\text{neutralizer}} H_2O$ + heat $\triangle H_3$

$\triangle H_1 + \triangle H_3 = \triangle H_2$

والذي نريد حساب

$\triangle H_3$

$\triangle H_{2(KJ/mol)}$ = - (4.18 X$\triangle$T X 102g)/ (1000 X moles of HCl (L.R))

Moles of HCl = M X V= 0.5 X 0.1=0.05

Moles heat of neutralization: $\triangle H_3 = \triangle H_2 - \triangle H_1$= -Ve

Questions:

1) When 6 g of NaOH (M.mass = 40 g/mol) dissolved in 90 g of water the temperature was increased from 22^o to 27^oC. if specific heat of solution = 4.18 J/g.°C and the density = 1 g/ml ΔH (KJ/mol) for solution:

A) -9.2 B) -7.1 C) -25.9 D) -13.4 Answer: **D**

Solution:

$\Delta H = -$ (ΔT X mass solution X Specific heat)/ (1000 X mole of NaOH)

$\Delta H = -$ (27-22) X (6+90) X 4.18) / (1000 X 0.15)

$\Delta H = -13.4$ KJ/mol.

2) When 5 g of NaOH dissolved in 100 g of diluted HCL solution the temperature was changed by 4^o C. if the same amount of NaOH dissolved in 100 g of water, the change in temperature will be:

A) Higher than 4 ^oC B) Lower than 4 ^oC

C) Equal 4 ^oC D) Equal zero

Answer: **B**

3) When 5.0 g of NaOH dissolved in 100 g of water, A 18.3 KJ of heat was evolved, if the same amount of NaOH dissolved in 100 g of HCl solution, the amount of heat evolved is:

A) equal 18.3 KJ B) Higher than 18.3 KJ

C) Lower than 18.3 KJ D) equal zero

Answer: **B**

4) The amount of heat evolved through the dissolving of 2 g NaOH in 100 g HCl solution was recorded to be 95 KJ/mol, if heat of neutralization equal 55 KJ/mol. The amount of heat evolved through the dissolving of 2 g NaOH in 100 g water is equal to:

A) 30 KJ/mol B) 140 KJ/mol C) 150 KJ/mol D) 40 KJ/mol

Answer: **D**

Solution:

$\Delta H_{(neutralization)} = \Delta H_{(NaOH\ and\ HCl)} - \Delta H_{(NaOH\ and\ water)}$

$55 = 95 - \Delta H_{(NaOH\ and\ water)}$

$\Delta H_{(NaOH\ and\ water)} = 40$ KJ/mol

5) How many grams of NaOH should be dissolved in 65 g of water to produce 1.3 KJ of heat? If the temperature changed from 20^{o} to 24^{o} C. (assume specific heat = 4.18 J/g.°C, and density = 1 g/ml)

A) 12.8g B) 18.7g C) 24.7g D) 6.8g Answer: **A**

Solution:

يجب تحويل وحده heat من KJ الى J لان وحده specific heat هي J

$\Delta H = - \Delta T$ X mass of solution X specific heat

1.3 X 1000 = - (4 X mass of solution X 4.18)

1300 = 16.72 mass of solution

Mass of solution = 77.75

Mass of solution = mass of water + mass of NaOH

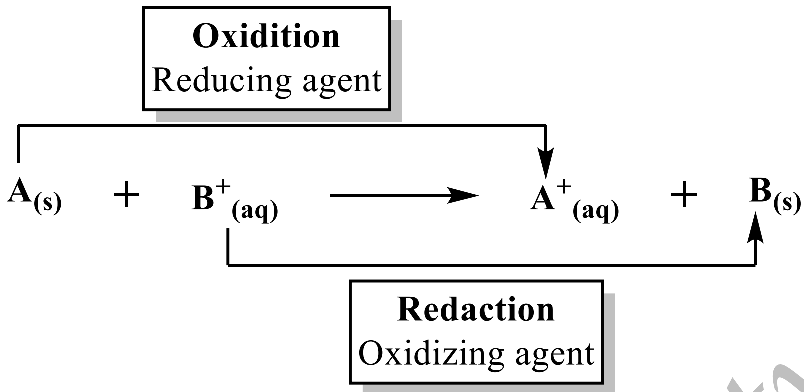
77.75 = 65 + mass of NaOH

Mass of NaOH = 12.8 g

Experiment number 7: Redox Reaction

Reduction: gaining of electron (oxidizing agent)

Oxidation: losing of electron (reducing agent)

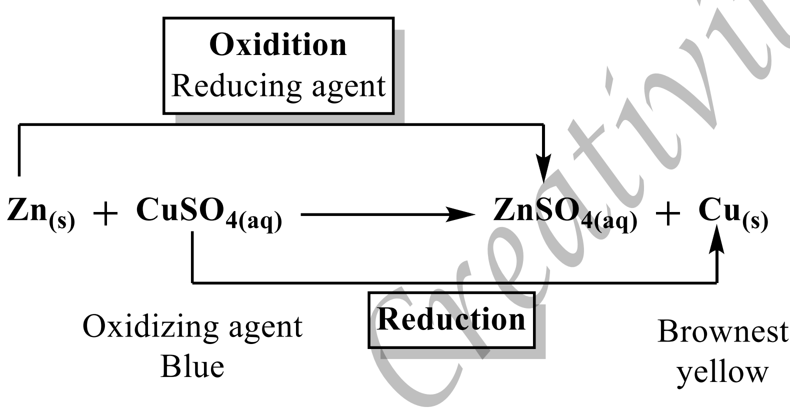


التاكسد و الاختزال: هما مترافقان في جميع التفاعلات

اي انه لا يمكن ان يحدث اختزال بدون تاكسد والعكس صحيح

الاختزال هو زيادة في الالكترونات ونقصان في الشحنة

التأكسد هو نقصان قي الالكترونات و زيادة في الشحنة



In Lab:

1. Add 10ml unknown (use pipet) conc. $CuSO_4$ solute and weigh.
2. Add 1g of Zn (excess).
3. Shaking for 10 min.
4. Add 15ml of 6 M HCl solution dissolve the unreacted (remained) Zn.
5. Filtration of Cu.
6. Drying Cu in oven 15 min, weighing Cu.

$$Zn_{(s)} + 2H^{+} + 2e^{-} \longrightarrow Zn^{+2} + 2H^{+} + 2e^{-}$$

كل تفاعلات التأكسد والاختزال

في أتجاه **واحد فقط**

$$Zn_{(s)} + 2HCl_{(aq)} \longrightarrow ZnCl_2^{1-} + H_{2(g)} \uparrow$$

$$Cu + 2HCl_{(aq)} \longrightarrow \text{No reaction}$$

All redox reaction is irreversible.

Calculation:

1. Mass of Cu= mass of Cu and filter paper after drying – mass filter paper
2. Moles of Cu= mass/molar mass=63.5
3. Moles of $CuSO_4$ = moles of Cu, ratio 1:1
4. Molarity of $CuSO_4$ = moles of Cu / Volume(L) $10X10^{-3}$
5. Mass of $CuSO_4$ solution = Volume X density =10 X 3.6 = 30.6g
6. Mass% of Cu = {mass of Cu /mass of solution (30.6)} X 100%

Questions:

1) How many grams of Ag (M.wt = 108 g/mol) produced from the redox reaction of 8.2 ml of 0.23 M $AgNO_3$ solution with excess of Cu metal to Oxidize it to Cu^{+2} :

A) 0.25g B) 0.4g C) 0.1g D) 0.2g <u>Answer: **D**</u>

Solution:

$$2AgNO_3 + Cu \longrightarrow 2Ag^+ + Cu(NO_3)_2$$

Mole of $AgNO_3$ = M X V

Mole of $AgNO_3$ = 0.23 X 8.2 X 10^{-3}

Mole of $AgNO_3$ = 1.886 X 10^{-3}

2 mol $AgNO_3 \longrightarrow$ 2 mol Ag

1.866 X $10^{-3} \longrightarrow$??????

Mole Ag = 1.866 X 10^{-3}

$$\text{Mole} = \frac{\text{Mass}}{\text{M.wt}} \longrightarrow \text{Mass} = \text{Mole X M.wt}$$

Mass = 1.866 X 10^{-3} X 108

Mass = 0.2 g

2) Consider the following redox reaction:

$$Zn + CdSO_4 \longrightarrow ZnSO_4 + Cd$$

If sufficient amount of Zn is added to 15 ml solution of $CdSO_4$ (M.wt = 208.4) having density of 1.07 g/ml and the mass of the precepted Cd is 3.542 g, then the mass% of Cd in $CdSO_4$ solution is:

A) 22.1% B) 15.8% C) 20.2% D) 14.6%

<u>Answer: **A**</u>

Solution:

$$d = \frac{\text{Mass}}{\text{Volume}}$$

$$1.07 \text{ g/ml} = \frac{\text{Mass}}{15 \text{ ml}} \longrightarrow \text{Mass} = 16.05 \text{ g}$$

$$\text{Cd\%} = \frac{\text{Actual}}{\text{Theoretical}} \text{ X } 100\%$$

$$\text{Cd\%} = \frac{3.542}{16.05} \text{ X } 100\%$$

Cd% = 22.1%

3) In the quantitive yield of redox reaction, which of the following was used to dissolve excess Zn unreacted:

A) HNO3 B) CuSO4 C) NaOH D) NH3 E) HCl

Answer: **E**

4) In the following reaction

$$\mathbf{A + B \longrightarrow A^{+} + B}$$

A) A is oxidizing agent and B^+ is reducing agent

B) A is reducing agent and B^+ is oxidizing agent

C) A^+ is oxidizing agent and B is reducing agent

D) A^+ is reducing agent and B is oxidizing agent

Answer: **B**

5) If 0.75 g of Cu (M.wt = 63 g/mol) produced from the reaction of 15 ml of $CuSO_4$ solution with excess Zn, the molarity of $CuSO_4$ solution is:

A) 0.59 B) 0.40 C) 0.48 D) 0.79 Answer: **D**

$$CuSO_4 + Zn \longrightarrow ZnSO_4 + Cu$$

Mole Cu = Mole $CuSO_4$

$$\frac{\text{Mass}}{\text{M.wt}} = M \times V$$

$$\frac{0.75}{63} = M \times 15 \times 10^{-3} \longrightarrow M = 0.79$$

6) The element that has ability to lose electrons is classified as:

A) Oxidized element and oxidizing agent

B) Reduced element and reducing agent

C) Oxidized element and reducing agent

D) Reduced element and oxidizing agent

Answer: **C**

Experiment number 8: Acid Base Titration

Purpose: determine the unknown concentration.

الهدف من التجربة ايجاد تركيز مجهول
من خلال المعايرة

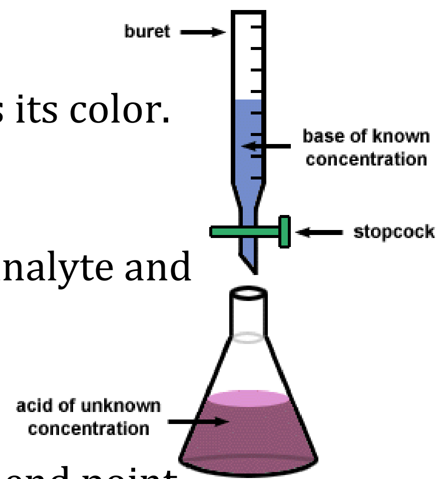
Titration: is a reaction between Known Conc. Substance (standard or titrant) with unknown's Conc. (analyte).

تفاعل بين مادتين أحدهما مجهول التركيز
والاخر معلوم التركيز

Indicator: weak organic acid or base change its color at the end point.

الكواشف: حمض او قاعدة ضعيفة,
الهدف منها تغيير اللون اثناء المعايرة

The indicator today: phenolphthalein (Ph.Ph), pink in basic medium and colorless in acidic medium.



End point: the point at which the indicator changes its color.

هذه النقطة هي نقطة النهاية التي يتغير عندها اللون

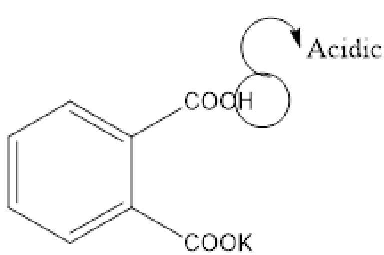
Equivalent point: the point at which the moles of analyte and standard are stoichiometry equal.

هذه النقطة هي نقطة المعايرة والتي تصبح فيها النسبة 1:1

Always equivalent point comes be or take place the end point.

In Lab:

Part A: standardization of NaOH with KHP determination of correct Conc.:



KHP: Potassium hydrogen phthalate Acid

KHP: Primary standard acid, 1-pure 2- stable with time 3-not hygroscopic 4-high molar mass.

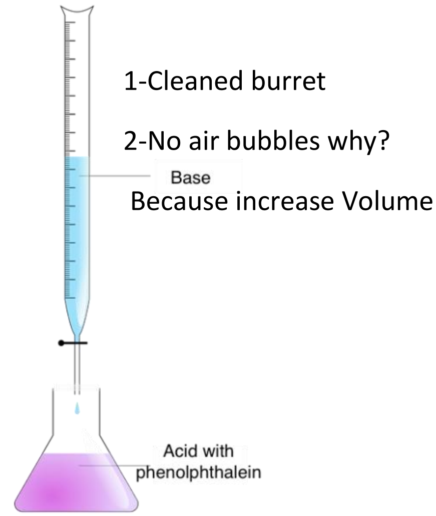
NaOH: Secondary standard base, hygroscopic.

1. 0.4 g of KHP
2. Dissolved in water
3. Add 2 or 3 drops of ph.ph.
4. Add NaOH from burret until

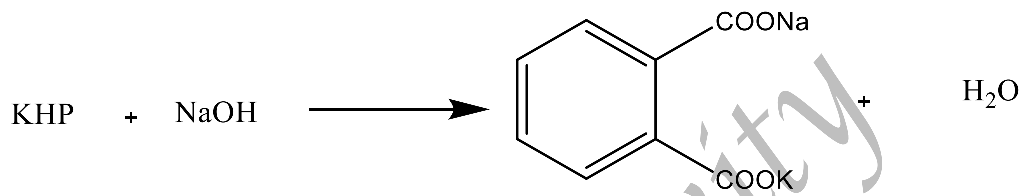
The pink color appears (end point)

5. Record of volume NaOH from burret

Volume $_{(NaOH)ML}$= final reading – initial reading



Calculation:



1-Moles of KHP= mass/molar mass=0.4/204.2

2-Moles NaOH= Moles of KHP

3-Molarity of NaOH= moles of NaOH/Volume (L) ml

Part B: titration of H_2SO_4 with NaOH:

1. In flask 10ml H_2SO_4
2. Use pipet
3. Add 2 or 3 drops of Ph.Ph. indicator
4. Add NaOH from burret until the pink color appear then record volume of NaOH.

Calculation:

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$, Na_2SO_4 diprotic acid.

1-moles of NaOH solution = molarity X volume (L)

2-moles of H_2SO_4 = ½ Moles of NaOH

3-Molarity of H_2SO_4 solution = moles of H_2SO_4/Volume (L)

Questions:

1) A 0.1044 g sample of an unknown **monoprotic** acid requires 22.1 ml of 0.05 M NaOH to reach the end point, the molar mass of this acid is:

A) 0.095 B) 10.7 C) 72 D) 94.5 Answer: **D**

Solution:

End point $\longrightarrow$ mole acid = mole base

$$\frac{\text{Mass}}{\text{M.mass}} = M \times V \longrightarrow \frac{0.1044}{\text{M.mass}} = 0.05 \times 22.1 \times 10^{-3}$$

M.mass = 94.5 g/mol

2) 0.567 g sample of unknown diprotic acid require 8 ml of 0.2 M NaOH to reach the end point, the molar mass of this acid is in g/mol:

A) 405 B) 709 C) 473 D) 567 Answer: **B**

Solution:

diprotic acid $\longrightarrow H_2A$

$2NaOH + H_2A \longrightarrow$ Product

$$\text{Mol of } H_2A = \frac{\text{mole of NaOH}}{2}$$

$$\frac{\text{Mass}}{\text{M.mass}} = \frac{M \times V}{2}$$

$$\frac{0.567}{\text{M.mass}} = \frac{0.2 \times 8 \times 10^{-3}}{2}$$

M.mass = 709

3) The color of phenolphthalein (Ph.Ph) indicators is:

A) Pink in NaOH solution

B) Pink in H_2SO_4 solution

C) Colorless in NaOH solution

D) Yellow in H_2SO_4 solution

E) Yellow in NaOH solution

Answer: **A**

4) The end point of a titration reaction is the point at which:

A) The color of indicator changes

B) The number of equivalents of titrant and analyte is equal

C) Both reactions are consumed

D) All of the above

Answer: **A**

5) A 24 ml NaOH solution needed 14 ml of 0.15 M H_2SO_4 to be neutralized, the molarity of NaOH is:

A) 0.18 M B) 0.51 M C) 0.13 M D) 0.044 M

Answer: **A**

Solution:

Neutralized $\longrightarrow$ **Mole acid =** $\frac{1}{2}$ **Mole base**

M X V = MX V

$0.15 \times 14 \times 10^{-3} = \frac{M \times 24 \times 10^{-3}}{2}$

M = 0.18

Experiment number 9: Rate Law

الهدف من التجربة ايجاد سرعة هذا التفاعل.

Purpose: determine the rate constant at a given temperature.

For the following reaction:

$$2I^-_{(aq)} + H_2O_{2(aq)} + 2H^+ \longrightarrow I_{2(aq)} + 2H_2O_{(L)}$$

هذا التفاعل سريع جدا" والوقت لا يمكن قراءته لذلك سوف نوضح كيف يصبح زمن مقروء.

$[H^+]$ constant by using buffer (PH=5) (HAc/Ac^-) if $[H^+]$ variated time very very low (fast reaction) and we can't obtain legible time.

قانون السرعة : وهو معدل اختفاء تركيز المتفاعلات و معدل ظهور النواتج مع الزمن.

$$\textbf{Rate} = \frac{-d[H_2O_2]}{dt} = \frac{-1\ d[I^-]}{2\ \ dt} = \frac{+d[I_2]}{dt} = \frac{+1\ d[H_2O]}{2\ \ dt} \quad \text{unit (mol/L.S)}$$

-ve = disappearance or consumption of reactant with time.

+ve = appear or production of products with time.

القانون الثاني : وهو كالتالي : أنظر لليسار .

Rate = K $[H_2O_2]^x$ $[I^-]^y$

حيث ان ال K هي ثابت السرعة

K: rate constant, X: order of H_2O_2, Y: order of I^-

X, Y وهما رتبة المواد المتفاعلة

Order: ratio between rate and initial concentration of reactants.

Order may be +ve, -ve, zero, integer, fraction.

و قد تكون اشارة موجبة او سالبة او صفر

+ve: as concentration for reactant increase the rate increase.

-ve: as concentration for reactant increase the rate decrease.

Zero: rate is not affected by the concentration.

In lab:

The rate of chemical reaction **depends on**:

1-**C**oncentration. 2-**T**empreture. 3-**S**urface area. 4-**C**atalyst.

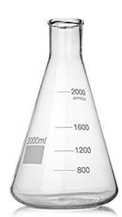
سرعة التفاعل تعتمد على:

التركيز , الحرارة ,

مساحة سطح التفاعل , العامل المحفز

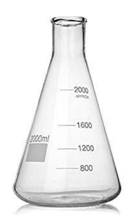
في الجزء العملي :

سوف نقوم بتحضير محلولين احدهما كالتالي :



water, KI, buffer, Starch, $Na_2S_2O_3$

Solution A



0.1 M H_2O_2

Solution B

Q: **Why** added water, KI, buffer, starch, $Na_2S_2O_3$?
Water: for dilution, small concentration to measure legible time.
Buffer: to stabilize $[H^+]$.
Starch: indicator.
$Na_2S_2O_3$: to consume the produced I_2 to give legible time.
KI, H_2O_2: main reactant.

$I_2 + 2S_2O_3^{-2} \longrightarrow 2I^- + S_4O_6^{-2}$ until all $S_2O_3^{-2}$ consumed completly, then:

$I_2 + \text{Starch} \longrightarrow \text{Dark blue color}$

In lab: when you add A (2, 3, 4) to B turn on the timer until the blue color appear record time (sec).

Calculation:

1. Moles of $Na_2S_2O_3$ = M X V(L) =0.02 X 5*10^{-3}
2. Moles of I_2 = ½ moles of $Na_2S_2O_3$
3. Rate = moles of I_2/time (s)
4. $$[I^-]_i = \frac{\overset{0.3}{[I^-]} \text{ X } V}{\underset{100ml}{V_{total}}}$$
5. $$[H_2O_2]_i = \frac{\overset{0.1}{[H_2O_2]} \text{ X } V}{\underset{100ml}{V_{total}}}$$

Rate = K $[H_2O_2]^x$ $[I^-]^y$ to calculate X, we stabilize $[I^-]$:

$$\frac{Rate_1 = K\,[H_2O_2]^x\,\cancel{[I^-]^y}}{Rate_2 = K\,[H_2O_2]^x\,\cancel{[I^-]^y}} \longrightarrow \frac{Rate_1 = K\,[H_2O_2]^x}{Rate_2 = K\,[H_2O_2]^x}$$

$$\text{Log }(rate_1/rate_2) = \text{X Log}\left(\frac{[H_2O_2]_1}{[H_2O_2]_2}\right)$$

$X = \log(R_1/R_2) / \log([H_2O_2]_1 / [H_2O_2]_2)$
$Y = \log(R_1/R_2) / \log([I^-]_1 / [I^-]_2)$

$Rate = K [H_2O_2]^x [I^-]^y$
$K = Rate / [H_2O_2]^x [I^-]^y$
$K' = \sum K / 3$

Rate constant **depends on** temperature (As **T** increase the rate constant increase)
Rate constant (K) **independent on** the concentration of reactants.

Questions:

1) In the rate of the reaction of H_2O_2 with KI, **buffer** solution was used:

A) To stabilize the concentration of hydrogen ion to be constant during the reaction.
B) To consume I_2 in a reasonable time for measuring the rate of the reaction.
C) As indicator change its color when the reaction is finished.
D) To dilute concentrations of H_2O_2 and KI solution.

<u>Answer: **A.**</u>

2) The rate constant for a chemical reaction decrease with:

A) Decreasing of temperature.
B) Decreasing of the initial concentration of reactant.
C) Increasing of temperature.
D) Increasing of the initial concentration of reactant.

<u>Answer: **A.**</u>

3) In the rate of reaction of H_2O_2 with KI, the dark blue color was appeared when:

A) Buffer solution was consumed completely.
B) Na_2SO_3 was consumed completely.
C) H_2O_2 was consumed completely.
D) KI was consumed completely.

<u>Answer: **B.**</u>

4) Given the following reaction:

$$NO + \frac{1}{2}\ O_2 \longrightarrow NO_2$$

[NO]	[O_2]	Initial rate
0.2	0.1	2.0×10^{-5}
0.4	0.2	8.0×10^{-5}
0.4	0.1	8.0×10^{-5}

The rate law is:

A) $R = K\,[NO]\,[O_2]^2$

B) $R = K\,[NO]^2\,[O_2]$

C) $R = K[O_2]^2$

D) $R = K\,[NO]\,[O_2]$

E) $R = K\,[NO]^2$

<u>Answer: **E**</u>

Solution:

➢ Rate = K $[NO]^x [O_2]^y$ to calculate Y, we stabilize [NO] in 2 and 3:

$$\frac{Rate_2 = K\,[NO]^x\,[O_2]^y}{Rate_3 = K\,[NO]^x\,[O_2]^y}$$

$$\frac{Rate_2 = K\,[0.4]^x[0.2]^y}{Rate_3 = K\,[0.4]^x[0.1]^y}$$

$$\frac{8 \times 10^{-3} \cancel{=} \cancel{K}\,\cancel{[0.4]}^x[0.2]^y}{8 \times 10^{-3} \cancel{=} \cancel{K}\,\cancel{[0.4]}^x[0.1]^y}$$

$$\text{Log } (8 \times 10^{-3}/8 \times 10^{-3}) = Y \text{ Log } \left(\frac{[0.2]}{[0.1]}\right)$$

$$\text{Log } 1 = Y \text{ Log } 2$$

$$Y = \frac{0}{0.3} = 0$$

➢ Rate = K $[NO]^x [O_2]^y$ to calculate X, we stabilize $[O_2]$ in 1 and 3:

$$\frac{Rate_1 = K\,[NO]^x\,[O_2]^y}{Rate_3 = K\,[NO]^x\,[O_2]^y}$$

$$\frac{Rate_2 = K\,[0.4]^x[0.1]^y}{Rate_3 = K\,[0.2]^x[0.1]^y}$$

$$\frac{8 \times 10^{-3} \cancel{=} \cancel{K}\,[0.4]^x\cancel{[0.1]^y}}{2 \times 10^{-3} = \cancel{K}\,[0.2]^x\cancel{[0.1]^y}}$$

$$\text{Log } (8 \times 10^{-3}/2 \times 10^{-3}) = X \text{ Log } \left(\frac{[0.4]}{[0.2]}\right)$$

$$\text{Log } 4 = X \text{ Log } 2$$

$$X = \frac{0.6}{0.3} = 2$$

Rate = K $[NO]^2$

Exp 10: Qualitative Analysis of cations

Objective: Separate a mixture of known cations Ag^+, $Pb^{2+}$and $Hg^{2+}{}_2$ into individual ions.

الهدف من التجربة كيفية فصل الايونات .

اولا" نقوم بأضافة حمض (HCL)

لترسيب الايونات

- **$Pb^{2+} + 2\,HCl \longrightarrow PbCl_{2(s)} + H^+$** white precipitate
- **$Ag^+ + HCl \longrightarrow AgCl_{(s)} + H^+$** white precipitate
- **$Hg^{2+}{}_2 + 2\,HCl \longrightarrow Hg_2Cl_{2(s)} + H^+$** white precipitate
- ✓ Chloride salts of these cations have low solubility in water – will ppt
- ✓ Add HCl will ppt out the salts and low acidity will prevent other solids from forming
- ✓ Can separate the 3 solids by variations in temperature or by forming complexes.

Begin with of the Known solution, which contains Ag^+, Hg^{2+} and Pb^{2+}, in a small test tube.

Ag^+, Hg^{2+}, Pb^{2+}

Add 6M HCl to the solution, to the precipitate that ions
Stir well. Separate with centrifuge.

Centerifuge

خطوة الترسيب

Water (supernatante)

White Ppt $AgCl(s)$, $PbCl_2(s)$,$Hg_2Cl_2(s)$

نقوم بالتخلص من الرائق ونضيف ماء ساخن لفصل Pb

Added hot water to dissolved $PbCl_2$

Centerifuge

Added $K_2Cr_2O_7$ (yellow solution)

Supernatant $(Pb^{2+})aq$

White ppt ($AgCl$,Hg_2Cl_2)

Yellow ppt ($PbCro_4$)

$Pb_2{}^+ + CrO_4{}^{2-} \longrightarrow PbCrO_4$

الباقي من الايونات نضيف عليهم NH3

Added 6 M NH_3

Centerifuge

اذا ظهر اللون الاسود او الرمادي يدل على وجود الزئبق

Black or grey
$Hg(L)$ black + $HgNH_2Cl(s)$ white

Supernatante
$Ag(NH_3)^{2+}Cl^-$

Added HNO_3 (Acid)

Soluble colorless complex

Ag^+ white ppt ($AgCl$)

$$AgCl(s) + 2NH_3\,(aq) \longrightarrow Ag(NH_3)^{2+} + Cl^- \longrightarrow AgCl(s) + 2NH_3NO_3\,(aq)$$

$$Hg_2Cl_2(s) + 2NH_3\,(aq) \longrightarrow Hg\,(l) + HgNH_2Cl + NH4Cl\,(aq)$$

نسحب الرائق ونضيف عليه HNO3

ونقوم بترسيب الفضة حسب المعادلات

Questions:

1) Unknown I was treated with six drops of HCl. A precipitate was observed. Which ion or group of ions was precipitate:

A) Pd B) Pt, Cu C) Ni D) Hg, Ag, and Pb Answer: **D**

2) The purpose of adding excess hot water to the ion solutions is:

A) Dissolve Ag only. B) Dissolve Hg and Ag.

C) Dissolve Pb only. D) Dissolve Pb and Hg.

Answer: **C.**

3) The purpose of adding excess 15 M NH3 to the test ion solutions is: (choose the single best answer)

A) Black precipitate of Hg. B) dissolve Ag to $Ag(NH_3)^{2+}Cl^{-1}$

C) dissolve Pb only. D) A+ B

Answer: **D**

4) After dissolve Pb with hot water and remove supernatant, we check the presence of lead by adding what:

A) Add 6M of NH_3 and precipitated $Pb(NH_3)_2$.

B) Add 6M of HCl and precipitated $PbCl_2$.

C) Add HNO_3

D) Add $K_2Cr_2O_7$ and precipitated $PbCrO_4$ (yellow ppt).

Answer: **D**

5) Which of the following equations are True and the false equations re-write to true:

I. $Pb^{2+} + 2\,HCl \longrightarrow PbCl_{2(s)} + H^+$

II. $Pb^{2+} + CrO_4^{2-} \longrightarrow PbCr_4O_{7(s)}$

III. $Hg_2Cl_{2(S)} + 2NH_{3(aq)} \longrightarrow Hg_{(L)} + HgNH_2Cl + NH_4Cl_{(aq)}$

IV. $AgCl_{(s)} + 2NH_{3\,(aq)} \longrightarrow Ag(NO_3) + Cl^-$